

First Terminal Examination - 2079

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternatives.

[1 × 11 = 11]

- Which of the following atoms is expected to have a larger atomic radius than others?
a. Boron b. Oxygen c. Nitrogen d. Fluorine
- The correct electronic configuration to represent d-block elements is,
a. $(n-1)d^{1-10} np^{1-6}$ b. $(n-1)d^{1-10} ns^{1-2}$
c. $nd^{1-10} ns^{1-2}$ d. $(n-2)d^{1-10} ns^{1-2}$
- The name chalcogen is given to which group of periodic table?
a. 15 b. 17 c. 16 d. 18
- Transition elements include,
a. Group 1-2 elements b. Group 13-18 elements
c. Group 3-12 elements d. Group 1-10 elements
- Which of the following statement is wrong about electron?
a. It is a particle
b. Its motion is affected by magnetic field
c. It has wave like property
d. It emits energy while moving in orbit.
- Who coined the word "atom"?
a. Democritus b. Thomson
c. Rutherford d. John Dalton

7. Which of the following determines the chemical properties?
 - a. Number of protons.
 - b. Number of electrons.
 - c. Number of neutrons.
 - d. Number of atom.
8. Which of the following isotope has equal number of proton and neutron?
 - a. ${}^1_1\text{H}$
 - b. ${}^2_1\text{H}$
 - c. ${}^3_1\text{H}$
 - d. ${}^7_3\text{Li}$
9. Which of the following particle is emitted by tritium?
 - a. α - particle
 - b. β - particle
 - c. x - ray
 - d. neutron
10. Boiling point of heavy water is
 - a. 100.42°C
 - b. 101.42°C
 - c. 104.20°C
 - d. 105.42°C
11. O_2^{2-} is the symbol of ion.
 - a. oxide
 - b. superoxide
 - c. peroxide
 - d. monoxide

Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. How can you justify the superiority of the modern periodic table over Mendeleev's periodic table with reference to, i) position of isotopes
ii) cause of periodicity iii) anomalous pairs? [2+2+1]
2. It is not possible to measure atomic radii due to various reasons. In what way the size of an atom and ions are expressed? Define one of them. Explain the periodic variation of atomic radii in groups and periods with reasons. [1+1+3]
3. Rutherford's nuclear model of atom is based on alpha particles scattering experiment.
 - a. Why did Rutherford select gold for his experiment? [1]
 - b. What conclusion can be drawn from Rutherford's alpha particle scattering experiment by observing back scattering of α -particles. Also, draw a diagram of experiment with proper labeling. [2]
 - c. What are the shortcomings of Rutherford's nuclear model of atom? [2]

4. Both Rutherford and Bohr cannot explain the complete structure of atom, so different model were developed to explain structure of atom.
- Explain de Broglie wave mechanical model of atom. [2]
 - What is the significance of de Broglie equation? [1]
 - Which observations of Rutherford's alpha ray scattering experiment conclude that atom consists of empty space and heavy positive charge? [2]
5. Bohr's atomic theory is able to explain the stability of atom as well as the formation of hydrogen spectra.
- Draw a labeled diagram to show different spectral series of hydrogen spectra. [2]
 - Explain the spectral line observed in visible and UV region. [2]
 - Write the difference between continuous and discontinuous spectra. [1]
6. Hydrogen and oxygen both are non -metals and are placed in separate groups in the periodic table.
- What is heavy water? Mention its uses. [2]
 - Define the term allotropy with examples. [1]
 - How does hydrogen resemble both alkali metals as well as halogens? [1]
 - Write two applications of atomic hydrogen. [1]
7. Oxides have been classified into different types on the basis of acidic, basic behaviour, structural arrangements etc.
- What type of oxide is Na_2O_2 ? Justify. [2]
 - Is Cl_2O an oxide? Give reason to your answer. [1]
 - Mention one example each of [1]
 - Neutral oxide
 - Acidic oxide
 - Write any two uses of tritium [1]

8. The applications of oxides are associated with their properties. The key properties are mechanical, electrical and magnetic, thermal, biomedical etc. Some of the oxides are given below,

ZnO	FeO	Mn ₃ O ₄
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- Classify the oxides from the above table on the basis of chemical behavior with the help of chemical reaction. [3]
- Mention the name and formula of the oxides (anhydrides) of following acids: HNO₃, H₃PO₄, H₂CrO₄ and H₂CO₃ [2]

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

- Explain the structural features of the modern periodic table. Write characteristics of d-block elements. [4+2]
 - How are the elements classified on the basis of the number of incomplete shells? Explain one of them. [2]
- Different model was put forward to explain the structure of atom, one of them is Rutherford's nuclear model of atom.
 - Which model improved it? Explain the postulates of the improved model. [4]
 - Write the shortcoming of improved model. [2]
 - Explain, why hydrogen shows large number of spectral lines though it contains single electron [2]
- The different forms of hydrogen are atomic hydrogen, nascent hydrogen, molecular hydrogen etc.
 - What is nascent hydrogen? Explain, with any two examples to show that nascent hydrogen is better reducing agent than molecular hydrogen. Explain with proper chemical reactions. [1+3]
 - What are the isotopes of hydrogen? Which is the least abundant isotope of hydrogen? Also, mention its percentage of abundance. [2]
 - Why is hydrogen called zero emission fuel? Write two applications of hydrogen as fuel. [2]

❧ - The End - ❧